~ Anomalous Zeeman effect

A very important theorem of quantum mechanics is the Wigner - Eckart theorem (see p. 238-242, Sakurai). This theorem allows one to know the matrix transition element of a tensor operator in the basis of angular-momentum eigenstates $|l s j m_j\rangle$. For the same j (and thus l & s) and a vectorial operator $\hat{\vec{V}}$, it simplifies to the so-called projection theorem:

$$\langle l s j m_j' | \hat{V}_q | l s j m_j \rangle = \frac{\langle l s j m_j | \hat{\vec{J}} \cdot \hat{\vec{V}} | l s j m_j \rangle}{\hbar^2 j (j+1)} \langle l s j m_j' | \hat{J}_q | l s j m_j \rangle ,$$

where the index q refers to the so-called spherical components of the ops.:

$$\hat{J}_{\pm 1} = \mp \frac{1}{\sqrt{2}} (\hat{J}_x \pm i \hat{J}_y) , \qquad \hat{J}_0 = \hat{J}_z .$$

We shall use this result in the following

~

As we already saw, ~~in the presence of~~ an electron carries two contributions to its magnetic dipole moment:

$$\vec{\mu}_l = -\frac{e}{2m} \vec{L} = -g_l \frac{\mu_B}{\hbar} \vec{L} , \qquad g_l = 1 \qquad \text{orbital motion}$$

$$\vec{\mu}_s = -\frac{e}{m} \vec{S} = -g_s \frac{\mu_B}{\hbar} \vec{S} , \qquad g_s = 2 \qquad \text{intrinsic spin}$$

with the Bohr magneton $\mu_B \equiv \dfrac{e\hbar}{2m}$.

Thus, the classical (normal) Zeeman effect must be replaced by using

$$\hat{V}_{\text{Zeeman}} = -\hat{\vec{\mu}} \cdot \vec{B} \qquad \hat{\vec{\mu}} = \hat{\vec{\mu}}_l + \hat{\vec{\mu}}_s = -\frac{\mu_B}{\hbar} (\hat{\vec{L}} + 2\hat{\vec{S}})$$

$$= -\frac{\mu_B}{\hbar} (\hat{\vec{J}} + \hat{\vec{S}})$$

By choosing $\vec{B} = B \vec{e}_z$, we have

$$\hat{V}_{\text{Zeeman}} = \frac{\mu_B B}{\hbar} (\hat{J}_z + \hat{S}_z) .$$

Although it's tempting to work in the basis $|l\, m_l\, s\, m_s\rangle$, we must remember that these states are not eigenstates of $\hat{H}$ if the spin-orbit term is included AND it <u>must</u> be included. Thus, in order to know the correction to the energies E_n of the eigenstates $|n\, l\, s\, j\, m_j\rangle$ due to the $\vec{B}$, we must compute

$$* \quad \langle \hat{V}_{Zeeman}\rangle = \langle n\, l\, s\, j\, m_j | \hat{V}_{Zeeman} | n\, l\, s\, j\, m_j\rangle = \langle l\, s\, j\, m_j | \hat{J}_z + \hat{S}_z | l\, s\, j\, m_j\rangle \frac{\mu_B B}{\hbar}$$

The first term is easy to compute while the second one requires using the Wigner-Eckart theorem:

$$\langle l s j m_j | \hat{S}_z | l s j m_j\rangle = \frac{\langle l s j m_j | \hat{\vec{J}}\cdot\hat{\vec{S}} | l s j m_j\rangle}{\hbar^2 j(j+1)} \langle l s j m_j | \hat{J}_z | l s j m_j\rangle$$

$$= \frac{\langle l s j m_j | \hat{\vec{L}}\cdot\hat{\vec{S}} + \hat{\vec{S}}^2 | l s j m_j\rangle}{\hbar^2 j(j+1)} m_j \hbar$$

$$= \frac{m_j}{\hbar j(j+1)} \frac{1}{2} \langle l s j m_j | \hat{\vec{J}}^2 - \hat{\vec{L}}^2 + \hat{\vec{S}}^2 | l s j m_j\rangle$$

$$= \frac{m_j \hbar}{2 j(j+1)} \left[j(j+1) - l(l+1) + s(s+1) \right]$$

$$* \quad \Rightarrow \boxed{\begin{aligned} \Delta E^{Zeeman}_{jlsm_j} &= \mu_B B m_j \Big[\underbrace{\frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)} + \overbrace{1}^{\langle \hat{J}_z\rangle}}_{\equiv g} \Big] \\ &\equiv \mu_B B \underbrace{g}_{\text{Landé } g\text{-factor}} m_j \end{aligned}}$$

For $s = 1/2 \rightsquigarrow j = l+s$ or $j = l-s$ $(l>0)$, it's easy to show that

$$g = \frac{j+1/2}{l+1/2}$$

For $l=0$, $\quad g = \dfrac{1/2 + 1/2}{1/2} = 2 = g_s$ for the electron spin

Note also that if $s=0 \rightarrow j=l \rightarrow g=1=g_l$ (classical result)

This equation for g allows further generalizations for arbitrary j.

~ Example: Lyman series

$n=2 \longrightarrow n'=1$

Selection rules: $2p \longrightarrow 1s$

$$1s: \ |n\, l\, s\, j\, m_j\rangle = \begin{cases} |1\,0\,\tfrac{1}{2}\,\tfrac{1}{2}\,+\tfrac{1}{2}\rangle \\ |1\,0\,\tfrac{1}{2}\,\tfrac{1}{2}\,-\tfrac{1}{2}\rangle \end{cases} \qquad j=1/2 \rightarrow 1s^{1/2},\ m_j=\pm\tfrac{1}{2}$$

$$2p: \ |n\, l\, s\, j\, m_j\rangle = \begin{cases} |2\,1\,\tfrac{1}{2}\,\tfrac{1}{2}\,+\tfrac{1}{2}\rangle \\ |2\,1\,\tfrac{1}{2}\,\tfrac{1}{2}\,-\tfrac{1}{2}\rangle \\ |2\,1\,\tfrac{1}{2}\,\tfrac{3}{2}\,+\tfrac{3}{2}\rangle \\ |2\,1\,\tfrac{1}{2}\,\tfrac{3}{2}\,\tfrac{1}{2}\rangle \\ |2\,1\,\tfrac{1}{2}\,\tfrac{3}{2}\,-\tfrac{1}{2}\rangle \\ |2\,1\,\tfrac{1}{2}\,\tfrac{3}{2}\,-\tfrac{3}{2}\rangle \end{cases} \qquad \begin{array}{l} j=1/2 \rightarrow 2p^{1/2},\ m_j=\pm\tfrac{1}{2} \\ \\ j=3/2 \rightarrow 2p^{3/2},\ m_j=\pm\tfrac{1}{2},\pm\tfrac{3}{2} \end{array}$$

$\cancel{\Delta E^{Zeeman} = \mu_B B}$

<u>1s</u>

$$\Delta E^{Zeeman}_{\frac{1}{2}0\frac{1}{2}\frac{1}{2}} = \mu_B B \frac{1}{2}\,\frac{1}{1/2} = \mu_B B$$

$$\Delta E^{Zeeman}_{\frac{1}{2}0\frac{1}{2}-\frac{1}{2}} = -\mu_B B$$

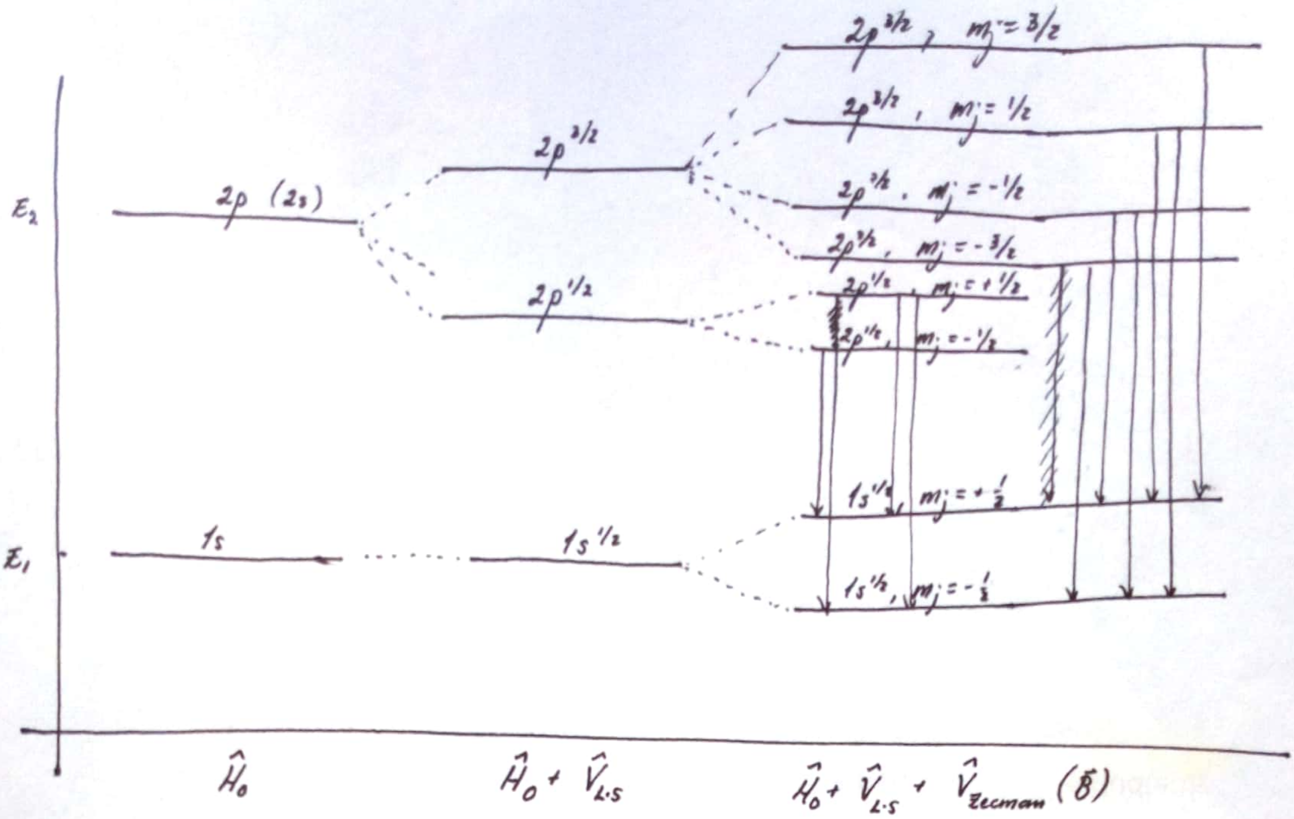
<u>2p</u>

$$\Delta E^{Zeeman}_{\frac{1}{2}1\frac{1}{2}\frac{1}{2}} = \mu_B B\, \frac{1}{2}\,\frac{1}{3/2} = \frac{1}{3}\mu_B B$$

$$\Delta E^{Zeeman}_{\frac{1}{2}1\frac{1}{2}-\frac{1}{2}} = -\frac{1}{3}\mu_B B$$

$$\Delta E_{\frac{3}{2}1\frac{1}{2}\pm\frac{3}{2}} = \pm\mu_B B\,\frac{3}{2}\,\frac{2}{3/2} = \pm 2\mu_B B$$

$$\Delta E_{3/2\,1\frac{1}{2}\pm\frac{1}{2}} = \pm\mu_B B\,\frac{1}{2}\,\frac{2}{3/2} = \pm\frac{2}{3}\mu_B B$$



In order to determine the possible transitions, it's needed to compute

$$\langle n'l's'j'm_j'|\hat{\vec{r}}|nlsjm_j\rangle$$

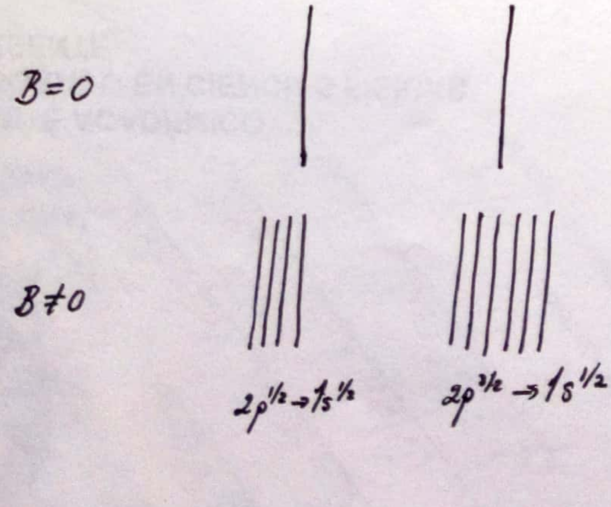
once again. However, ~~since~~ it's easy to see that, since $\hat{\vec{r}}$ doesn't affect the spin status &

$$m_j = m_l + m_s ,$$

only transitions with $(\Delta m_l = 0, \pm 1)$

$$\Delta m_j = 0, \pm 1$$

are allowed. That's why



§ 6.3. More complex systems: the Pauli's principle

The hydrogen atom (and hydrogen-like atoms such as those of the first family in the periodic table) is the simplest quantum system. In general, it's not possible to reduce the quantum systems to problems of one-particle dynamics. Thus, we are forced to consider systems with multiple particles, which can be in different states.

Let's consider multiparticle systems, such that

$$\hat{H} = \sum_i \hat{H}_i , \qquad i=1,2,3,\dots \text{ counts particles } \left(\Rightarrow E = \sum_i E_i\right)$$

i.e. for which there's no interacting potential among the different particles. In such a case, we can denote the state of the system as

$$|N_1 N_2 N_3 \dots\rangle \equiv |N_1\rangle |N_2\rangle |N_3\rangle | \dots , \qquad \text{(energy eigenstates)}$$

such that N_i is the set of quantum numbers of the i-th particle. E.g. for the ~~hydrogen~~ atoms with hydrogen-like quantum numbers

$$N_i = \{ n_i, l_i, s_i, j_i, m_{j_i} \} \quad \text{for each electron}$$

We shall also use the representation in the position space of these states

$$\psi_{N_i}(\vec{r}_i) \equiv \langle \vec{r}_i | N_i \rangle$$

$$\Rightarrow \quad \langle \vec{r} | N_1 N_2 \dots \rangle \equiv \langle \vec{r}_1 | N_1 \rangle \langle \vec{r}_2 | N_2 \rangle \cdots$$

$$= \psi_{N_1}(\vec{r}_1)\, \psi_{N_2}(\vec{r}_2) \cdots \equiv \psi(\vec{r}_1, \vec{r}_2, \dots)$$

~ Identical particles

The simplest multiparticle case is for identical particles. This is particularly interesting when studying the ~~atomic~~ electronic structure of atoms or the structure of pure molecules, for which the identical particles are electrons and atoms, respectively.

In those cases, we shall have in general

$$\vec{L} = \sum_i \vec{L}_i ,$$
$$\vec{S} = \sum_i \vec{S}_i$$
$$\& \quad \vec{J} = \vec{L} + \vec{S} .$$

The eigenvalues of these angular momenta are restricted by the same rules we had before. However, it's important to notice that e.g. two electrons ($s_i = 1/2$) do not only yield $\underline{s=1}$ but also $\underline{s=0}$. The precise addition rules shall be studied elsewhere.

The important observation is that we may have systems with identical half-integral-spin particles (such as the electrons in an arbitrary atom) or identical integral-spin particles (such as photons, α-particles, etc.). That is, we can consider either fermions or bosons.

Let's now define the permutation operator $\hat{P}_{ij}$ which exchanges the ~~quantum numbers~~ ~~of~~ i-th particle by the j-th particle. ~~Eg.~~ in the system.

$$\cancel{\hat{P}_{12} |N_1 N_2 N_3 \ldots \rangle = |N_2 N_1 N_3}$$

Since the particles are identical, we expect that

$$\hat{H} \doteq \hat{P}_{ij}^{-1} \hat{H} \hat{P}_{ij}$$

$$\Rightarrow \hat{P}_{ij} \hat{H} = \hat{H} \hat{P}_{ij} \qquad \text{i.e.} \qquad \boxed{[\hat{P}_{ij}, \hat{H}] = 0 \Rightarrow \frac{d\hat{P}_{ij}}{dt} = 0}$$

wrong! This ~~implies~~ that the energy eigenstates $|N_1 N_2 \ldots \rangle$ are also ~~eigenstates~~ of $\hat{P}_{ij}$, i.e.

BUT This is not an eigen-relation (we must write the eigenstate ... see later)

$$\hat{P}_{ij} |N_1 N_2 \ldots \rangle = \lambda_{ij} |N_1 N_2 \ldots \rangle = \lambda_{ij} |N_1 N_2 \ldots N_j N_i \ldots \rangle$$

$$\& \quad \hat{P}_{ij}^2 |N_1 N_2 \ldots \rangle = \lambda_{ij}^2 |N_1 N_2 \ldots \rangle = |N_1 N_2 \ldots \rangle = |N_1 N_2 \ldots N_i N_j \ldots \rangle$$

$$\lambda_{ij}^2 = 1 \quad \Rightarrow \quad \lambda_{ij} = \pm 1$$

$[\hat{P}_{ij}, \hat{H}]=0$ implies that we can find simultaneous eigenstates of $\hat{H}$ & $\hat{P}_{ij}$.

- First, we notice that $|N_1 N_2 \dots\rangle$ & $|N_2 N_1 \dots\rangle$ are eigenstates of $\hat{H}$ & that, for identical particles, they're undistinguishable.
- Second, we see that

$$\hat{P}_{ij} |\dots N_i \dots N_j \dots\rangle \longrightarrow |\dots N_j \dots N_i \dots\rangle$$

such that there may be a phase that distinguish the original & final states

From these observations, we propose that the eigenstates of $\hat{H}$ & $\hat{P}_{ij}$ are linear combinations

$$|\psi_{ij}\rangle = \Big[|\dots N_i \dots N_j \dots\rangle + \alpha_{ij} |\dots N_j \dots N_i \dots\rangle\Big] \underbrace{C_{ij}}_{\text{normalization}}$$

such that

$$\hat{P}_{ij} |\psi_{ij}\rangle = \lambda_{ij} |\psi_{ij}\rangle$$

$$\hat{H} |\psi_{ij}\rangle = \sum_{k=1}^{g} E_k |\psi_{ij}\rangle \equiv E |\psi_{ij}\rangle$$

Since $\hat{P}_{ij}^2 = \mathbb{1}$

$$\Rightarrow \quad \hat{P}_{ij}^2 |\psi_{ij}\rangle = \lambda_{ij}^2 |\psi_{ij}\rangle \overset{!}{=} |\psi_{ij}\rangle \quad \Rightarrow \quad \lambda_{ij} = \pm 1$$

However, these $|\psi_{ij}\rangle$ are not eigenstates of $\hat{P}_{kl}$ because, in general,

$$[\hat{P}_{ij}, \hat{P}_{kl}] \neq 0$$

e.g. in a 3-particle system

$$\hat{P}_{13} \hat{P}_{12} |N_1 N_2 N_3\rangle \longrightarrow \hat{P}_{13} |N_2 N_1 N_3\rangle \longrightarrow |N_2 N_3 N_1\rangle$$

$$\neq$$

$$\hat{P}_{12} \hat{P}_{13} |N_1 N_2 N_3\rangle \longrightarrow \hat{P}_{12} |N_3 N_2 N_1\rangle \longrightarrow |N_3 N_1 N_2\rangle$$

Our goal is to look for $\hat{H}$-eigenstates $|\psi\rangle$ that are eigenstates of all permutation ops. simultaneously wrong

~~However, in general,~~

$$\cancel{\left[\hat P_{ij}\,,\,\hat P_{jk}\right] \neq 0}$$

~~E.g. in the 3-particle system~~

$$\cancel{\hat P_{13}\,\hat P_{12}\,|N_1 N_2 N_3\rangle = \lambda_{12}\,\hat P_{13}\,|N_2 N_1 N_3\rangle = \lambda_{12}\lambda_{13}\,|N_2 N_3 N_1\rangle}$$

$$\cancel{\hat P_{12}\,\hat P_{13}\,|N_1 N_2 N_3\rangle = \lambda_{13}\,\hat P_{12}\,|N_3 N_2 N_1\rangle = \lambda_{12}\lambda_{13}\,|N_3 N_1 N_2\rangle}$$

~~Thus, an arbitrary state $|N_1 N_2 \ldots\rangle$ is not necessarily eigenstate of $\hat P_{ij}$ & $\hat P_{jk}$ simultaneously!~~ We can, ~~however~~ build, 2 exceptional cases:

1) Symmetric $\quad |S\rangle = \overset{\text{normalization}}{C_s} \displaystyle\sum_{\text{perms.}} \hat P_{ij}\,|N_1 N_2 \ldots\rangle$

2) Antisymmetric $\quad |A\rangle = C_A \displaystyle\sum_{\text{perms}} (-1)^{r_p}\,\hat P_{ij}\,|N_1 N_2 \ldots\rangle \qquad r_p = \begin{cases} +1 & \text{odd perm.} \\ 0 & \text{even perm} \end{cases}$

The states $|S\rangle$ & $|A\rangle$ are more physical than simply $|N_1 N_2 \ldots\rangle$ for identical particles, because it's in practice impossible to know whether the system is in the state

$$|N_1 N_2 \ldots\rangle$$

$$\text{or} \quad |N_2 N_1 \ldots\rangle$$

for example. That is, the state is in a superposition of them. However the relative signs could in principle be arbitrary. These states fulfill $\hat P_{ij}|S\rangle = +|S\rangle$, $\hat P_{ij}|A\rangle = -|A\rangle \quad \forall\, i,j$.

Pauli observed, however, in 1925 that the electronic structure of atoms in the periodic table could only be explained iff

a) Identical particles with <u>integral spin</u> build <u>symmetric states $|S\rangle$</u> (bosons)

b) " " " <u>half-integral spin</u> " <u>antisymmetric states $|A\rangle$</u> (fermions)

No other combination is observed in nature!

2-particle example:

$$|S\rangle = \frac{1}{\sqrt{2}}\Big(|N_1N_2\rangle + |N_2N_1\rangle\Big) = \frac{1}{\sqrt{2}}\Big(|N_1\rangle|N_2\rangle + |N_2\rangle|N_1\rangle\Big)$$

$$|A\rangle = \frac{1}{\sqrt{2}}\Big(|N_1N_2\rangle - |N_2N_1\rangle\Big) = \frac{1}{\sqrt{2}}\Big(|N_1\rangle|N_2\rangle - |N_2\rangle|N_1\rangle\Big)$$

Suppose now that both particles are in the same state, $N_1 = N_2$

$$\Rightarrow \quad |S\rangle = \frac{1}{\sqrt{2}}\Big(2\,|N_2\rangle|N_2\rangle\Big) = \sqrt{2}\,|N\rangle|N\rangle$$

$$|A\rangle = \frac{1}{\sqrt{2}}\Big(|N\rangle|N\rangle - |N\rangle|N\rangle\Big) = 0 \qquad !!$$

[We find that bosons can be all found in the same state whereas fermions must be in different states! This is the so-called <u>Pauli's exclusion principle</u>.]

This 2-particle example can be readily generalized. Notice that $|A\rangle$ can be rewritten as

$\longrightarrow$ position of $|N_i\rangle$ in the perm.

$$|A\rangle = C \begin{vmatrix} |N_1\rangle_1 & |N_2\rangle_1 & \cdots \\ |N_1\rangle_2 & |N_2\rangle_2 & \cdots \\ |N_1\rangle_3 & |N_2\rangle_3 & \cdots \\ \vdots & \vdots & \end{vmatrix} \quad \Big\downarrow \text{state}$$

called the Slater determinant. It's clear that if $N_i = N_j$ for any i,j, $|A\rangle$ cancels.

Later, it was found that fermions with ~~…~~ states $|A\rangle$ build energy/temp density distributions described by the so-called Fermi-Dirac distribution

$$\rho = \frac{\overset{\text{degeneracy of state}}{2}}{e^{(E-E_F)/kT}+1} \qquad E_F:\ \text{Fermi energy}$$

whereas bosons build a Bose-Einstein distribution

$$\rho = \frac{\overset{\text{degeneracy}}{1}}{e^{(E-E_0)/kT}-1} \qquad E_0:\ \text{ground state energy.}$$

For $E \gg kT$, both yield $\rho \sim e^{-E/kT}$ Boltzmann's distribution (classical)

§6.4 Density Matrix or density operator

Usually, a state is represented by $|\psi\rangle$, which may be a <u>superposition</u>.

This is called a <u>PURE state</u> and represents a situation in which the whole system is in a state, despite of being composed of ~~subst~~ subsystems.

A system may be composed of subsystems which are found in different states. A system is said to be in a <u>statistical ensemble</u> or <u>MIXED state</u> if e.g. 50% of the system is found in state $|\psi_1\rangle$ & 50% in state $|\psi_2\rangle$.

These are <u>classical</u> probabilities in contrast to the <u>quantum</u> probs of a superposition.

An ensemble in which the fraction p_j is in state $|j\rangle$ is described by the density operator

$$\hat{\rho} = \sum_j p_j |j\rangle\langle j| \quad \ni \quad p_j \geq 0 \quad \& \quad \sum_j p_j = 1$$

We note that for a <u>PURE state</u> in the state $|i\rangle$

$$\hat{\rho} = |i\rangle\langle i|$$

The states $\{|j\rangle\}$ may <u>not</u> be orthogonal!

We can ~~compute~~ define the expectation value of an op. $\hat{A}$ in the ensemble as

$$\langle \hat{A} \rangle \equiv \sum_j p_j A_j \quad , \quad A_j = \langle j|\hat{A}|j\rangle$$

We note that this can be rewritten, if $|j\rangle = \sum_n c_{jn} |n\rangle$ in complete, orthonormal basis $\{|n\rangle\}$,
$= \sum_n \langle n|j\rangle |n\rangle$

as
$$\langle \hat{A} \rangle = \sum_j p_j \langle j|\hat{A}|j\rangle = \sum_j \sum_{m,n} p_j c_{jm}^* \langle m|\hat{A}|n\rangle c_{jn}$$

$$= \sum_j \sum_{m,n} p_j \langle j|m\rangle \langle m|\hat{A}|n\rangle \langle n|j\rangle$$

$$= \sum_{m,n} \langle m|\hat{A}|n\rangle \langle n| \left(\sum_j p_j |j\rangle\langle j| \right) |m\rangle$$

$$= \sum_{m,n} \langle m|\hat{A}|n\rangle \langle n|\hat{\rho}|m\rangle = \sum_m \langle m|(\hat{A}\hat{\rho})|m\rangle$$

$$= tr(\hat{A}\hat{\rho}) = tr(\hat{\rho}\hat{A})$$

$$\Rightarrow \boxed{\langle \hat{A} \rangle = tr(\hat{\rho}\hat{A})}$$

~ Props. of $\hat{\rho}$

- If $\hat{A} = \hat{\mathbb{1}} \Rightarrow \langle \hat{A} \rangle = 1$

$$\Rightarrow \boxed{tr(\hat{\rho}) = 1}$$

Also verified from

$$tr(\hat{\rho}) = \sum_m \sum_j p_j \langle m|j\rangle\langle j|m\rangle = \sum_j p_j \langle j| \left(\sum_m |m\rangle\langle m|\right) |j\rangle$$
$$= \sum_j p_j \langle j|j\rangle = \sum_j p_j = 1$$

- Self-adjoint or Hermitian

$$\hat{\rho}^\dagger = \left(\sum_j p_j |j\rangle\langle j|\right)^\dagger = \sum_j p_j^* |j\rangle\langle j| = \sum_j p_j |j\rangle\langle j| = \hat{\rho} \quad p_j \in \mathbb{R}$$

$$\boxed{\hat{\rho}^\dagger = \hat{\rho}}$$

- Positive-definite

Since $p_j \geq 0 \Rightarrow \langle \psi | \hat{\rho} | \psi \rangle \geq 0$

- In a pure state $\hat{\rho} = |i\rangle\langle i|$

$$\hat{\rho}^2 = (|i\rangle\langle i|)(|i\rangle\langle i|) = |i\rangle\langle i|i\rangle\langle i| = |i\rangle\langle i| = \hat{\rho}$$

$$\Rightarrow \boxed{\hat{\rho}^2 = \hat{\rho}}$$

~ Example: electron-spin mixture

A (pure) state can be written as

$$|\psi\rangle = a|+\rangle + b|-\rangle \quad \to \quad |a|^2 + |b|^2 = 1$$

We observe that, in this basis, the pure-state density matrix is

$$\hat{\rho} = |\psi\rangle\langle\psi| = (a|+\rangle + b|-\rangle)(a^*\langle +| + b^*\langle -|)$$
$$= aa^* |+\rangle\langle +| + ab^* |+\rangle\langle -| + ba^* |-\rangle\langle +| + bb^* |-\rangle\langle -|$$

$$\Rightarrow \hat{\rho} = \begin{pmatrix} aa^* & ab^* \\ ba^* & bb^* \end{pmatrix} \quad \text{in the basis } \{|+\rangle, |-\rangle\}$$

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Note that the could use another basis, in which $|\varphi\rangle$ and

$$|\varphi'\rangle = c|+\rangle + d|-\rangle , \qquad |c|^2+|d|^2=1,$$

such that $\langle\varphi|\varphi'\rangle = 0$, were the basis vectors. In such a basis, the density matrix would read

$$\hat{\rho} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

and would be equivalent to the previous one. We learn that different density matrices yield the same physical description.

Turning back to the $\{|+\rangle, |-\rangle\}$ basis, we may entertain some special cases:

$\hat{\rho}_+ = |+\rangle\langle+| \Rightarrow$ electrons are fully polarized with $m_s = +\frac{1}{2}$

$\hat{\rho}_- = |-\rangle\langle-| \Rightarrow$ " " " " " $m_s = -\frac{1}{2}$

$$\hat{\rho}_x = \left(\frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle\right)\left(\frac{1}{\sqrt{2}}\langle+| + \frac{1}{\sqrt{2}}\langle-|\right)$$

$$= \frac{1}{2}\left(|+\rangle\langle+| + |+\rangle\langle-| + |-\rangle\langle+| + |-\rangle\langle-|\right)$$ polarized in the x-direction

In all these cases $\hat{\rho}^2 = \hat{\rho} \rightarrow$ pure state.

BUT We can build mixed states such as

$$\hat{\rho} = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix} = p|+\rangle\langle+| + (1-p)|-\rangle\langle-|$$

corresponding to a mixture of a fraction of p particles in the state $|+\rangle$ and a fraction of $1-p$ particles in the state $|-\rangle$. We note that in this case

$$\hat{\rho}^2 = \begin{pmatrix} p^2 & \\ & (1-p)^2 \end{pmatrix} \neq \hat{\rho} \quad \text{if } p \neq 0,1$$

~ Electron polarization

We know that

$$\hat{\vec{S}} = \frac{1}{2}\hbar\hat{\vec{\sigma}}$$

In an ensemble of electrons, the measurement of ~~all possible~~ the spin corresponds to what we typically call polarization of an electron beam

$$\Rightarrow \quad \frac{1}{2}\hbar\vec{P} \equiv \langle\hat{\vec{S}}\rangle = \frac{1}{2}\hbar\langle\hat{\vec{\sigma}}\rangle,$$

where $\vec{P} \equiv \langle\hat{\vec{\sigma}}\rangle$ is the polarization vector.

$$\Rightarrow \quad \vec{P} = tr(\hat{\rho}\hat{\vec{\sigma}})$$

On the other hand, $\hat{\rho}$ can be written as the combination

$$\hat{\rho} = \sum_{\mu}\alpha_{\mu}\hat{\sigma}_{\mu} \qquad \mu = 0,\dots,3 \quad \rightarrow \quad \hat{\sigma}_0 = \mathbb{1}_2$$

Since $tr\,\hat{\sigma}_i = 0$ for $i = 1,2,3$, and $tr\,\hat{\sigma}_0 = tr\,\mathbb{1}_2 = 2$

$$\Rightarrow \quad tr\hat{\rho} = 2\alpha_0 \doteq 1 \qquad \Rightarrow \quad \alpha_0 = \frac{1}{2}$$

Let's compute now the component

$$P_i = tr(\hat{\rho}\,\hat{\sigma}_i) = tr\left[\left(\frac{1}{2}\mathbb{1}_2 + \sum_j\alpha_j\hat{\sigma}_j\right)\hat{\sigma}_i\right]$$

$$= \sum_j tr\left[\alpha_j\hat{\sigma}_j\hat{\sigma}_i\right] = \sum_j\alpha_j\, tr\left[\hat{\sigma}_j\hat{\sigma}_i\right]$$

~~Since $\hat{\sigma}_j\hat{\sigma}_i = \delta_{ij}\mathbb{1}_2 + i\epsilon_{jik}\hat{\sigma}_k$~~ $\qquad \hat{\sigma}_j\hat{\sigma}_i = \delta_{ij}\mathbb{1}_2 + i\epsilon_{jik}\hat{\sigma}_k$

~~$\Rightarrow P_i$~~

Since $tr\left[\hat{\sigma}_j\hat{\sigma}_i\right] = 2\delta_{ji} \quad \Rightarrow \quad P_i = 2\alpha_i \quad \Rightarrow \quad \alpha_i = \frac{1}{2}P_i$

$$\Rightarrow \quad \hat{\rho} = \frac{1}{2}\left(\mathbb{1}_2 + \vec{P}\cdot\hat{\vec{\sigma}}\right) \doteq \frac{1}{2}\begin{pmatrix} 1+P_3 & P_1 - iP_2 \\ P_1 + iP_2 & 1-P_3 \end{pmatrix}$$

We can compare to $\hat{\rho}$ ~~obtained~~ obtained for the pure state $a|+\rangle + b|-\rangle$

$$\begin{pmatrix} aa^* & ab^* \\ ba^* & bb^* \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1+P_3 & P_1 - iP_2 \\ P_1 + iP_2 & 1-P_3 \end{pmatrix}$$

$$\Rightarrow \quad |a|^2 = \tfrac{1}{2}(1+P_3) \qquad \Rightarrow \qquad a = \sqrt{\tfrac{1}{2}(1+P_3)}\; e^{i\alpha}$$

$$|b|^2 = \tfrac{1}{2}(1-P_3) \qquad\qquad b = \sqrt{\tfrac{1}{2}(1-P_3)}\; e^{i\beta}$$

$$\cancel{ab^* = \tfrac{1}{2}\sqrt{(1-P_3^2)}\; e^{i(\alpha-\beta)} = \tfrac{1}{2}(P_1 - iP_2)}$$

$$\Rightarrow \cancel{\sqrt{1-P_3^2}\,\left(\cos(\alpha-\beta) + i\,\mathrm{sen}(\alpha-\beta)\right) = P_1 - iP_2}$$

$$\cancel{ba^* = \tfrac{1}{2}\sqrt{1-P_3^2}\; e^{\,i}}$$

$$ab^* = \tfrac{1}{2}\sqrt{1-P_3^2}\; e^{i(\alpha-\beta)} = \tfrac{1}{2}(P_1 - iP_2)$$

$$\sqrt{1-P_3^2}\,\left(\cos(\alpha-\beta) + i\,\mathrm{sen}(\beta-\alpha)\right) = P_1 - iP_2$$

$$\Rightarrow \quad \tan(\beta-\alpha) = \frac{P_2}{P_1}$$

or vice versa

$$P_3 = 2|a|^2 - 1\,, \qquad P_1 = ab^* + ba^*\,, \qquad P_2 = -i(ba^* - ab^*)$$

$$\Rightarrow \vec{P}\cdot\vec{P} = \cancel{(ab^*)^2} + \cancel{(ba^*)^2} + 2|a|^2|b|^2 - \cancel{(ab^*)^2} - \cancel{(ba^*)^2} + 2|a|^2|b|^2 + 4|a|^4 + 1 - 4|a|^2$$

$$= 4|a|^2|b|^2 + 4|a|^2(1-|b|^2) + 1 - 4|a|^2$$

$$= 1$$

$$\Rightarrow \quad \vec{P}\cdot\vec{P} = 1 \quad \Rightarrow$$ A pure state of spin ½ is totally polarized in a direction

For example, if $a=1, b=0$, $|\psi\rangle = |+\rangle$

$$\Rightarrow P_1=0,\ P_2=0,\ P_3=+1 \rightarrow \text{polarized beam in dir. } S_3^+$$

If $a=0, b=1$, $|\psi\rangle = |-\rangle$

$$\Rightarrow P_1=0,\ P_2=0,\ P_3=1-2|b|^2=-1 \rightarrow \text{polarized beam in dir } S_3^-$$

If $a=\frac{1}{\sqrt{2}}, b=\frac{1}{\sqrt{2}}$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|+\rangle+|-\rangle)$

$$\Rightarrow P_1=1,\ P_2=0,\ P_3=0 \rightarrow \text{polarized beam in dir. } S_1^+$$

as we can show

$$\hat{S}_1|\psi\rangle \doteq \frac{1}{2}\hbar\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \left(\frac{1}{2}\hbar\right)\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{2}\hbar|\psi\rangle$$

If $a=\frac{1}{\sqrt{2}}, b=-i\frac{1}{\sqrt{2}}$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|+\rangle - i|-\rangle)$

$$P_1=0,\ P_2=-i(-i+i)=-1,\ P_3=0 \rightarrow \text{polarized beam in dir } S_2^-$$

matching our expectation because

$$\hat{S}_2|\psi\rangle = \frac{1}{2}\hbar\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix} = -\frac{1}{2}\hbar|\psi\rangle$$

Now, for a mixed state of the form $\hat{\rho} = p|+\rangle\langle+| + (1-p)|-\rangle\langle-|$

we have

$$\begin{pmatrix}p & \\ & 1-p\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1+P_3 & P_1-iP_2\\ P_1+iP_2 & 1-P_3\end{pmatrix}$$

$$\Rightarrow P_1=P_2=0,\quad P_3=2p-1 \rightsquigarrow \text{only partly polarized in } S_3$$

$$\Rightarrow \vec{P}\cdot\vec{P} = (2p-1)^2 \neq 1 \quad \text{for } p\neq 0,1$$

i.e. the beam is not fully polarized. In fact, if $p=\frac{1}{2}$, the beam is <u>not polarized</u> at all, for $\vec{P}\cdot\vec{P}=0$, which is compatible with the idea of having 50% of ~~states in~~ electrons in state $|+\rangle$ and 50% in state $|-\rangle$.

~ Bosons vs. fermions

If we describe particles as harmonic oscillators,

$$\Rightarrow \quad \hat{H} = \hbar\omega\, \hat{a}^{+}\hat{a}$$

omitting the zero-point energy contribution $\frac{1}{2}\hbar\omega$ for convenience.

As we have seen, $[\hat{a}, \hat{a}^{+}] = 1$. This only applies to <u>bosonic</u> oscillators.

If the oscillators are <u>fermionic</u>, we must demand rather

$$\{\hat{a}, \hat{a}^{+}\} = \hat{a}\hat{a}^{+} + \hat{a}^{+}\hat{a} = 1$$

According to Planck's hypothesis, in the case of photons (bosons), the fraction of the photons with energy E_n is

$$\rho_n = \frac{1}{Z} e^{-\beta E_n}, \quad \beta = k_B T, \quad Z = \sum_n e^{-\beta E_n}$$

With this info

$$\hat{\rho} = \frac{1}{Z} \sum_n e^{-\beta E_n} |n\rangle\langle n|$$

$$= \frac{1}{Z} \sum_n e^{-\beta \hat{H}} |n\rangle\langle n|$$

$$= \frac{1}{Z} e^{-\beta \hat{H}} \sum_n |n\rangle\langle n| \quad = \frac{1}{Z} e^{-\beta \hat{H}}$$

We can rewrite

$$Z = \mathrm{tr}\, e^{-\beta \hat{H}} = \sum_n \langle n| e^{-\beta \hat{H}} |n\rangle = \sum_n e^{-\beta E_n}$$

$$\Rightarrow \quad \hat{\rho} = \frac{e^{-\beta \hat{H}}}{\mathrm{tr}\, e^{-\beta \hat{H}}}$$

~~If we compute the expectation value of the energy, we obtain~~

$$\cancel{\langle E \rangle = \mathrm{tr}(\rho \hat{H}) = \hbar\omega \frac{\sum_n \langle n| \hat{a}^{+}\hat{a}\, e^{-\beta \hat{a}^{+}\hat{a}\hbar\omega} |n\rangle}{\sum_n \langle n| e^{-\beta\hbar\omega \hat{a}^{+}\hat{a}} |n\rangle}}$$

$$\cancel{= \hbar\omega \frac{\sum_n n\, e^{-\beta\hbar\omega n}}{\sum_n e^{-\beta\hbar\omega n}} = -\frac{\partial}{\partial\beta} \ln Z}$$

Notice that

$$\langle E\rangle = tr(\rho\hat{H}) = \frac{tr(e^{-\beta\hat{H}}\hat{H})}{tr(e^{-\beta\hat{H}})} = \frac{tr(e^{-\beta\hat{H}}\hat{H})}{Z} = -\frac{\partial}{\partial\beta}\ln Z$$

Now, taking the value of the energies in a bosonic oscillator, we obtain

$$Z = \sum_n e^{-\beta E_n} = \sum_n (e^{-\beta\hbar\omega})^n,$$

which for $e^{-\beta\hbar\omega} < 1$ leads to the geometric series

$$Z = \frac{1}{1-e^{-\beta\hbar\omega}} = \frac{e^{\beta\hbar\omega}}{e^{\beta\hbar\omega}-1}$$

$$\Rightarrow \langle E\rangle = -\frac{\partial}{\partial\beta}\ln\left(\frac{1}{1-e^{-\beta\hbar\omega}}\right) = \frac{\partial}{\partial\beta}\ln(1-e^{-\beta\hbar\omega}) = \frac{\hbar\omega\, e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}$$

$$= \frac{\hbar\omega}{e^{\beta\hbar\omega}-1}.$$

Another method to arrive at this result and generalize it is as follows. Consider

$$tr(e^{-\beta\hat{H}}\hat{H}) = \hbar\omega\, tr(e^{-\beta\hat{H}}\hat{a}^{+}\hat{a}) = \hbar\omega\, tr(\hat{a}\, e^{-\beta\hat{H}}\hat{a}^{+}).$$

Now, we use the following (bosonic) results

$$[\hat{a},\hat{H}] = \hbar\omega a, \qquad [\hat{a},\hat{H}^2] = [\hat{a},\hat{H}]\hat{H} + \hat{H}[\hat{a},\hat{H}] = (\hbar\omega)^2 a + 2\hbar\omega H a \qquad (*)$$

$$[\hat{a},\hat{H}^3] = [\hat{a},\hat{H}^2]\hat{H} + \hat{H}^2[\hat{a},\hat{H}] = (\hbar\omega)^3\hat{a} + 3(\hbar\omega)^2\hat{H}\hat{a} + 3\hbar\omega\hat{H}^2\hat{a}$$

to observe that

$$\hat{a}e^{-\beta\hat{H}} = \hat{a} - \beta\hat{a}\hat{H} + \frac{\beta^2}{2}\hat{a}\hat{H}^2 - \frac{\beta^3}{3!}\hat{a}\hat{H}^3 + \dots$$

$$= \hat{a} - \beta(\hbar\omega\hat{a} + \hat{H}\hat{a}) + \frac{\beta^2}{2}\left[(\hbar\omega)^2\hat{a} + 2\hbar\omega\hat{H}\hat{a} + \hat{H}^2\hat{a}\right] - \frac{\beta^3}{3!}\left[(\hbar\omega)^3\hat{a} + 3(\hbar\omega)^2\hat{H}\hat{a} + 3\hbar\omega\hat{H}^2\hat{a} + \hat{H}^3\hat{a}\right] + \dots$$

$$= a - \beta\hbar\omega\hat{a} + \frac{\beta^2}{2}(\hbar\omega)^2\hat{a} - \frac{\beta^3}{3!}(\hbar\omega)^3\hat{a} - \beta\hat{H}a + \beta^2\hbar\omega\hat{H}\hat{a} - \frac{\beta^3}{2}(\hbar\omega)^2\hat{H}\hat{a} + \frac{\beta^2}{2}\hat{H}^2 a - \frac{\beta^3}{2}\hbar\omega\hat{H}^2\hat{a}$$

$$= \left(1 - \beta\hbar\omega + \frac{\beta^2}{2}(\hbar\omega)^2 - \frac{\beta^3}{3!}(\hbar\omega)^3 + \dots\right)\left(1 - \beta\hat{H} + \frac{\beta^2}{2}\hat{H}^2 + \dots\right)\hat{a}$$

$$= e^{-\beta\hbar\omega}e^{-\beta\hat{H}}\hat{a}$$

This implies that

$$\mathrm{tr}\left(\hat{a}\, e^{-\beta \hat{H}} \hat{a}^{+}\right) = e^{-\beta\hbar\omega}\, \mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}\hat{a}^{+}\right) = e^{-\beta\hbar\omega}\, \mathrm{tr}\left(e^{-\beta\hat{H}}(1+\hat{a}^{+}\hat{a})\right)$$

$$= e^{-\beta\hbar\omega}\left(Z + \mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}^{+}a\right)\right).$$

We can generalize this result to <u>fermions</u>, by replacing the commutators in p. 103 by anticommutators. This doesn't change the last result of p. 103, but the previous result becomes

$$\mathrm{tr}\left(\hat{a}\, e^{-\beta\hat{H}}\hat{a}^{+}\right) = e^{-\beta\hbar\omega}\, \mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}\hat{a}^{+}\right) = e^{-\beta\hbar\omega}\, \mathrm{tr}\left(e^{-\beta\hat{H}}(1-\hat{a}^{+}\hat{a})\right)$$

because of $\{\hat{a}, \hat{a}^{+}\} = 1$. That is, we have in general that

$$\mathrm{tr}\left(\hat{a}\, e^{-\beta\hat{H}}\hat{a}^{+}\right) = e^{-\beta\hbar\omega}\left(Z \pm \mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}^{+}\hat{a}\right)\right),$$

depending on whether we have bosonic (+) or fermionic (-) oscillators. Recalling that $\mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}^{+}\hat{a}\right) = \mathrm{tr}\left(\hat{a}\, e^{-\beta\hat{H}}\hat{a}^{+}\right)$, we find

$$\left(1 \mp e^{-\beta\hbar\omega}\right)\mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}^{+}\hat{a}\right) = e^{-\beta\hbar\omega} Z$$

It follows that the expectation value of the energy reads

$$\langle E\rangle = \frac{\hbar\omega}{Z}\,\mathrm{tr}\left(e^{-\beta\hat{H}}\hat{a}^{+}\hat{a}\right) = \frac{\hbar\omega}{\cancel{Z}}\,\frac{e^{-\beta\hbar\omega}\cancel{Z}}{1\mp e^{-\beta\hbar\omega}} = \frac{\hbar\omega}{e^{+\beta\hbar\omega}\mp 1},$$

where the $-$ signs (Bose-Einstein statistics) correspond to bosons and the $+$ (Fermi-Dirac statistics) to fermions. By using

$$\langle E\rangle = -\frac{\partial}{\partial\beta}\ln Z,$$

we see that

$$Z_{bosons} = \frac{1}{1-e^{-\beta\hbar\omega}}$$

$$Z_{fermions} = 1 + e^{-\beta\hbar\omega}.$$

- Time evolution of $\hat{\rho}$

Since

$$|\psi(t)\rangle = \hat{\mathcal{U}}(t,0)|\psi_0\rangle\,,\quad |\psi_0\rangle \equiv |\psi(0)\rangle\,,\quad \hat{\mathcal{U}}\text{: time evol. op.}$$

and $i\hbar \frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle$

$$\Rightarrow \quad i\hbar \frac{\partial}{\partial t}\mathcal{U}(t,0) = \hat{H}\hat{\mathcal{U}}(t,0) \quad \text{Schrödinger's eq. for } \hat{\mathcal{U}}$$

It follows that

$$\hat{\rho}(t) = \sum_i p_i |i(t)\rangle\langle i(t)| = \sum_i p_i \hat{\mathcal{U}}(t,0)|i_0\rangle\langle i_0|\hat{\mathcal{U}}^\dagger(t,0)$$

$$= \hat{\mathcal{U}}(t,0)\left(\sum_i p_i |i_0\rangle\langle i_0|\right)\hat{\mathcal{U}}^\dagger(t,0)$$

$$= \hat{\mathcal{U}}(t,0)\,\hat{\rho}_0\,\hat{\mathcal{U}}^\dagger(t,0)$$

$$\Rightarrow \quad \frac{\partial\hat{\rho}}{\partial t} = \frac{\partial\hat{\mathcal{U}}}{\partial t}\hat{\rho}_0\hat{\mathcal{U}}^\dagger + \hat{\mathcal{U}}\hat{\rho}_0\frac{\partial\hat{\mathcal{U}}^\dagger}{\partial t} = \frac{1}{i\hbar}\left(\hat{H}\hat{\mathcal{U}}\hat{\rho}_0\hat{\mathcal{U}}^\dagger - \hat{\mathcal{U}}\hat{\rho}_0\hat{\mathcal{U}}^\dagger\hat{H}\right)$$

$$= \frac{1}{i\hbar}[\hat{H},\hat{\rho}]$$

$$\Rightarrow \quad \boxed{i\hbar\frac{\partial\hat{\rho}}{\partial t} = [\hat{H},\hat{\rho}]} \quad \text{ec. von Neumann time evol. of } \hat{\rho}$$

$$\Rightarrow \quad i\hbar\frac{\partial\hat{\rho}}{\partial t} + [\hat{\rho},\hat{H}] = 0$$

Recalling Heisenberg's eq. for $\hat{A} = \hat{A}(t)$

$$i\hbar\frac{d\hat{A}}{dt} = [\hat{A},\hat{H}] + i\hbar\frac{\partial\hat{A}}{\partial t}$$

This means that

$$\boxed{i\hbar\frac{d\hat{\rho}}{dt} = 0}$$

$\Rightarrow$ the number of states in a region is conserved!